Before the Public Service Commission

THE BROOKLYN UNION GAS COMPANY d/b/a NATIONAL GRID NY

And

KEYSPAN GAS EAST CORPORATION d/b/a NATIONAL GRID

Corrections and Updates Testimony

of

Climate Leadership and Community Protection Act Panel

Dated: June 30, 2023

I. Introduction

Q. Please identify the members of the Climate Leadership and Community Protection Act ("CLCPA") Panel ("Panel").

A. The Panel consists of Karsten A. Barde, Patricia J. Dorsch, and Meghan McGuinness.

Q. Is this the same Panel that previously submitted direct testimony on behalf of The Brooklyn Union Gas Company d/b/a National Grid NY (KEDNY") and KeySpan Gas East Corporation d/b/a National Grid ("KEDLI"((collectively, the "Companies") in these proceedings?

A. Yes. The defined terms in the Panel's direct testimony have the same meanings in this corrections and updates testimony.

Q. What is the purpose of the Panel's corrections and updates testimony?

A. The Panel's testimony describes the effects relevant to New York State's Climate Leadership and Community Protection Act ("CLCPA") related to updates in the Companies' investment plans proposed in these rate cases, as well as a correction identified in the course of responding to discovery. Specifically, the Panel describes CLCPA-related considerations related to updates to the Companies' respective electric vehicle ("EV") facilities and fleet forecasts, as described in the corrections and updates testimony of

each Company's Gas Infrastructure and Operations Panel ("GIOP"). In addition, KEDLI has revised the estimated greenhouse gas ("GHG") emissions projections of the proposed HyGrid project in Hempstead, New York. The impact of these changes is projected cumulative greenhouse gas ("GHG") emissions reductions for KEDNY of at least 873,700 metric tons of carbon dioxide equivalent (MT CO_2e) and for KEDLI of at least 778,000 MT CO_2e over the proposed four-year rate plan presented in the filings. The revised emissions estimates are approximately 1.4 percent less CO_2e reduction for KEDNY and approximately 0.9 percent less CO_2e reduction for KEDLI over the course of the proposed rate plan than the Panel projected in its direct testimony.

Q. Is the Panel sponsoring any exhibits as part of its corrections and updates filing?

A. Yes. The Panel sponsors the following exhibits that were prepared by one or more members of the Panel or under their supervision and direction:

(i) Exhibit___(CLCPA-3CU), Schedule 1: Revised Assessment of KEDNY Estimated Greenhouse Gas Emissions Impacts; Rate Year – Data Year 3.

(ii) Exhibit___(CLCPA-4CU), Schedule 1: Revised Assessment of KEDLI Estimated Greenhouse Gas Emissions Impacts; Rate Year – Data Year 3.

Additionally, KEDLI's response to Information Request ("IR") No. AGREE-08 (NG Request No. NG-427), which is relevant to the HyGrid project correction the Panel discusses below, is included in Appendix A to our testimony.

II. Corrections and Updates

Q. Please describe the projected GHG emissions effects associated with changes to the Companies' proposed fleet electrification plans.

A. As described in KEDNY's GIOP Corrections and Updates Testimony, KEDNY has revised downward the number of EVs it intends to replace over the course of the proposed rate plan. Similarly, as described in KEDLI's GIOP Corrections and Updates Testimony, KEDLI also has reduced the number of EVs it intends to replace over the course of the proposed rate plan. The Service Company also intends to reduce the number of EVs it acquires during the proposed rate plan, which in turn reduces the Service Company EV allocation to both KEDNY and KEDLI, respectively.

In addition to updating the fleet replacement schedules, the Companies' projected GHG emissions have been updated from the initial filing to incorporate historical data specific to each Company's fleet rather than averages based on fleet data from all National Grid operating companies. This is intended to provide a more accurate estimate of the GHG emissions benefit associated with electrifying the Companies' fleets. The updated information also corrects a cell reference error that had an immaterial effect on the emissions calculations.

Based on these corrections and updates, the Panel projects EV-related GHG emissions reductions at the end of Data Year 3 to be approximately 914 MT CO_2e for KEDNY and approximately 473 MT CO_2e for KEDLI. Exhibit__(CLCPA-3CU), Schedule 1, reflects the contribution of the revised EV-related GHG emissions reductions for KEDNY, and Exhibit__(CLCPA-4CU), Schedule 1, reflects the contribution of the revised EV-related GHG emissions reductions for KEDLI.

The Companies also are proposing to reduce the number of EV charger deployments at their facilities; however, the vast majority of charger deployments the Companies plan will be located in or adjacent to Disadvantaged Communities. Thus, the benefits of converting portions of

the Companies' fleets to cleaner and quieter operating alternatives will benefit those communities.

Q. Please describe the correction associated with the proposed HyGrid project.

A. In preparing its response to IR No. AGREE-08 (NG Request No. NG-427), KEDLI identified an error in its GHG emissions reduction calculation for the HyGrid project presented in Exhibit__(CLCPA-4). The revised projected GHG emissions reduction for the proposed HyGrid project is 518 MT CO_2e annually once placed in service. Exhibit__(CLCPA-4CU), Schedule 1, reflects the revised GHG emissions reductions for KEDLI with the HyGrid projected corrected.

Q. Does the schedule change for the LNG Tank 2 Foundation Heaters project described in KEDNY's GIOP Corrections and Updates Testimony affect the CLCPA Panel's projected emissions from that project?

A. No. The schedule change planned for the LNG Tank 2 Foundation Heaters project does not change the proposed in-service year of the project (FY 2028, or Data Year 3). Therefore, the updated schedule does not affect the associated GHG emissions effect.

Q. Do the changes described in KEDLI's GIOP Corrections and Updates Testimony related to the Holtsville LNG Plant affect the CLCPA Panel's projected emissions?

A. No. The corrections and update presented by the KEDLI GIOP associated with the Holtsville LNG Plant do not affect the projection of GHG emissions associated with that modernization project.

Q. Does the gas load forecast presented in the Corrections and Updates Testimony of the Gas Load Forecasting Panel affect the Companies' proposed billed gas usage reduction goal?

A. Yes. The Companies will seek to achieve a weather-adjusted, one-half of one percent reduction in forecast levels of gas usage (excluding non-core and electric generation) as included in the Corrections and Updates Testimony of the Gas Load Forecasting Panel Exhibit ___ (GLF-4ACU) for KEDNY and Exhibit ___ (GLF-4BCU) for KEDLI. The Companies propose to report on progress relative to these goals on an annual basis.

Q. Do other items identified in the Companies corrections and updates filings require changes in the Panel's testimony or exhibits?

A. No. The Panel has reviewed the items identified as part of this corrections and updates filing and determined that other than the items identified

above, no other changes presented in this filing would materially affect the Companies' projections of GHG emissions or effects on disadvantaged communities.

As with the Companies' initial rate case filings, the corrections and updates filings also comport with the requirements of the CLCPA.

Q. Does this conclude the Panel's testimony?

A. Yes.

Corrections and Updates Testimony of
Climate Leadership and Community Protection Act Panel

LIST OF EXHIBITS

Exhibit __ (CLCPA-3CU), Schedule 1:	Revised Assessment of KEDNY Estimated Greenhouse Gas Emissions Impacts; Rate Year – Data Year 3
Exhibit __ (CLCPA-4CU), Schedule 1:	Revised Assessment of KEDLI Estimated Greenhouse Gas Emissions Impacts; Rate Year – Data Year 3
APPENDIX A	Company Responses to Information Requests That Discuss Corrections and Updates- Information Request No. AGREE-08

Exhibit __ (CLCPA-3CU)
Schedule 1: Revised Assessment of KEDNY Estimated Greenhouse Gas Emissions Impacts; Rate Year – Data Year 3

Exhibit __ (CLCPA-3CU)
Schedule 1

KEDNY Emissions Reductions in Metric Tons of CO2e						
Workstream	**Initiatives**	**Rate Year**	**Data Year 1**	**Data Year 2**	**Data Year 3**	**Cumulative Emissions Rate Year-Date Year 3**
GIOP	Large Diameter Pipe Rehabilitation	740	1,481	2,221	2,962	7,404
	Large Diameter Pipe Rehabilitation - CISBOT	926	1,851	2,777	3,702	9,255
	Main Replacement - LPP - Proactive, CSC, Reliability	13,237	27,398	42,174	57,258	140,068
	Replace Pipe on Bridges	5	10	16	21	52
	Service Replacement - Atmospheric Corrosion Inside Inspections (Remediation)	665	1,331	1,996	2,662	6,654
	CNG fill station retirement - Brooklyn - Greenpoint				1,401	1,401
	CNG fill station retirement - Brooklyn - Canarsie				52	52
	LNG - Tail Gas Compressor Upgrade				-6	-6
	LNG - Relocate Maint. Area & New Control Building				-40	-40
	LNG - Tank 2 Foundation Heaters				-55	-55
	LNG - Salt Water Pump House Upgrade				-557	-557
	RNG Project - KEDNY 1 (avoided upstream emissions only)				6,103	6,103
	RNG Project - KEDNY 2 (avoided upstream emissions only)				22,490	22,490
	Fleet and Facilities	172	588	669	976	2,405
Customer	Energy Efficiency Programs	67,857	135,714	203,571	271,428	678,571
Total		83,603	168,373	253,424	368,397	873,797

List of Acronyms
CI: Cast Iron
CNG: Certified Natural Gas
CO2e: Carbon Dioxide Equivalent
CSC: City State Construction
LNG: Liquid Natural Gas
LPP: Leak Prone Pipe
PS: Protected Steel
RNG: Renewable Natural Gas
UPS: Unprotected Steel

Exhibit __ (CLCPA-4CU)
Schedule 1: Revised Assessment of KEDLI Estimated Greenhouse Gas Emissions Impacts;
Rate Year – Data Year 3

Exhibit __ (CLCPA-4CU)
Schedule 1

KEDLI Emissions Reductions in Metric Tons of CO2e						
Workstream	**Initiatives**	**Rate Year 1**	**Data Year 1**	**Data Year 2**	**Data Year 3**	**Cumulative Emissions Rate Year-Date Year 3**
GIOP	Large Diameter Pipe Rehabilitation	173	345	518	691	1,727
	LPP - Proactive, CSC, Reinforcement, Reliability	19,861	40,412	61,136	82,205	203,614
	Replace Pipe on Bridges	5	10	16	21	52
	CNG fill station retirement - Riverhead				52	52
	LNG - Holtsville Plant Modernization				-2,289	-2,289
	RNG Project - KEDLI 1 (avoided upstream emissions only)				4,841	4,841
	RNG Project - KEDLI 2 (avoided upstream emissions only)				4,284	4,284
	HyGrid				518	518
	Fleet and Facilities	294	455	615	749	2,113
Customer	Energy Efficiency Programs	57,128	114,256	171,384	228,512	571,281
Total		77,461	155,479	233,669	319,584	786,192

List of Acronyms

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CSC: City State Construction
LNG: Liquid Natural Gas
LPP: Leak Prone Pipe
PS: Protected Steel
RNG: Renewable Natural Gas
UPS: Unprotected Steel

Appendix A
Company Responses to Information Requests That Discuss Corrections and Updates
Information Request No. AGREE-08

Date of Request: June 5, 2023 Request No. AGREE- 08
Due Date: June 15, 2023 NG Request No. NG-427

KeySpan Gas East Corporation d/b/a National Grid
The Brooklyn Union Gas Company d/b/a National Grid NY

KEDNY KEDLI Case 23-G-0225/G-0226 Data Requests

Request for Information

FROM: Benjamin Kuebrich

TO: National Grid

SUBJECT: Gas Infrastructure and Operations Panel

Request:

In all interrogatories, all requests for workpapers or supporting calculations shall be construed as requesting any Word, Excel or other computer spreadsheet models in original electronic format with all formulae intact and unlocked.

The KEDLI GIOP panel describes hydrogen production and delivery to 844 customers in the Point Lookout section of Hempstead, NY.

a. What greenhouse gas emission reductions are projected with this project? Please provide the methodology involved in calculating the emissions reductions and please include all greenhouse gases, not just carbon reductions.

b. How much will this project cost in terms of dollars per ton of greenhouse gas emissions reduced?

c. Has the company compared the projected costs of GHG emission reductions to the cost of GHG emissions that could be achieved through full electrification for the 844 customers in the Point Lookout section of Hempstead, NY?

Response:

a. The Direct Testimony of Climate Leadership and Community Protection Act Panel provides the Companies' GHG analysis methodology. The equations and emission factors used to calculate greenhouse gas emissions were sourced from the Energy Sector Greenhouse Gas Emission Under the New York State Climate Act: 1990-2020, Table 25 and 26. The estimated GHG reductions for the HyGrid demonstration project are 518 MT annually based on the 20% blend rate. In preparing this response, the Company identified an inadvertent error in the initial greenhouse gas emissions reduction calculation originally provided in Exhibit ___ (CLCPA-4). The Company will reflect this revision in its Corrections and Updates filing.

b. The project is a proof-of-concept demonstration project and, therefore, the costs are not representative of a fully developed market and the potential for federal incentives to reduce costs. The estimated cost of the demonstration project based on the current pricing of electrolysis and related components is $13.291 million and assumes an electrolysis cost of $1,742 per/kW plus the cost to install which is indicative of low volume production.

The U.S. Department of Energy ("DOE") has adopted the Hydrogen Shot initiative and established a technology target of $1 per/kg of hydrogen in one decade. *See* https://www.energy.gov/eere/fuelcells/hydrogen-shot. In addition, DOE set a target of $300 per/kW for electrolysis by 2025 and $150 per/kW by 2030. If the HyGrid project were priced at targeted 2025 prices, the total project cost would be less than $7 million or $1,350 capital per metric ton of greenhouse gas emissions reductions over ten years at 20% blending before tax credits. In addition, the HyGrid project is expected to be eligible for a Production Tax Credit of $3 per/kg, or approximately $140,000 per/year, which has a twenty-year present value of $2.0 million.

c. As mentioned, the HyGrid project is a proof-of-concept demonstration project on a local scale and, therefore, it would not be instructive to compare the estimated per-unit costs of this project to electrification costs; however, future cost comparisons could be developed based on learnings from these types of demonstration projects. The Point Lookout neighborhood was developed in the 1920s as a community of summer bungalows and has since developed into full-year residences heated primarily with gas hydronic or steam heating systems. Multiple estimates have shown that the cost to retrofit a hydronically heated single-family residence with an air-source heat pump in this region can exceed $9,275 per home and be as high as $32,800 when including required weatherization to accommodate the air-source heat pumps. *See* Pathways to Carbon-Neutral NYC, at page 43, *available at* the NYC.gov website. The cost of a ground source system would be much higher and is not included for comparison purposes.

Assuming each of the 844 gas customers simultaneously replaced their natural gas heating system with a minimally compliant air-source heat pump, and assuming no building envelope improvements, the Company estimates customers would collectively incur an equipment cost of at least $7.8 million excluding costs of weatherization and appliance replacement. If the costs of weatherization needed to accommodate the heat pump are included, the Company estimates customer-sided costs increase to $27.68 million. These costs do not include any associated electric distribution system upgrades that may be required, and a complete comparison would require a detailed analysis of those upgrades and their estimated costs. The associated greenhouse gas emissions reductions and relative costs will be primarily dependent on the broader and quicker availability of renewable power than the HyGrid project requires and will also be dependent on future economies of scale.